温州肯恩大学
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GMBA 5885 W03
Strategic Mgt Global Info Sys

Building Information Systems

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Kick-Off Discussion

- Why do organizations need new information systems?

  Think of a situation where a business needed to improve a process, solve a problem, or adapt to change.

- What makes building a system difficult?

  Requirements? Data? Integration? Technology? Users? Which do you think is the biggest challenge?

- Build or Buy?

  If you were a manager, when would you build a custom system vs. buy SaaS or an application package?

- Development Approaches

  Have you heard of Agile, DevOps, or low-code/no-code tools?

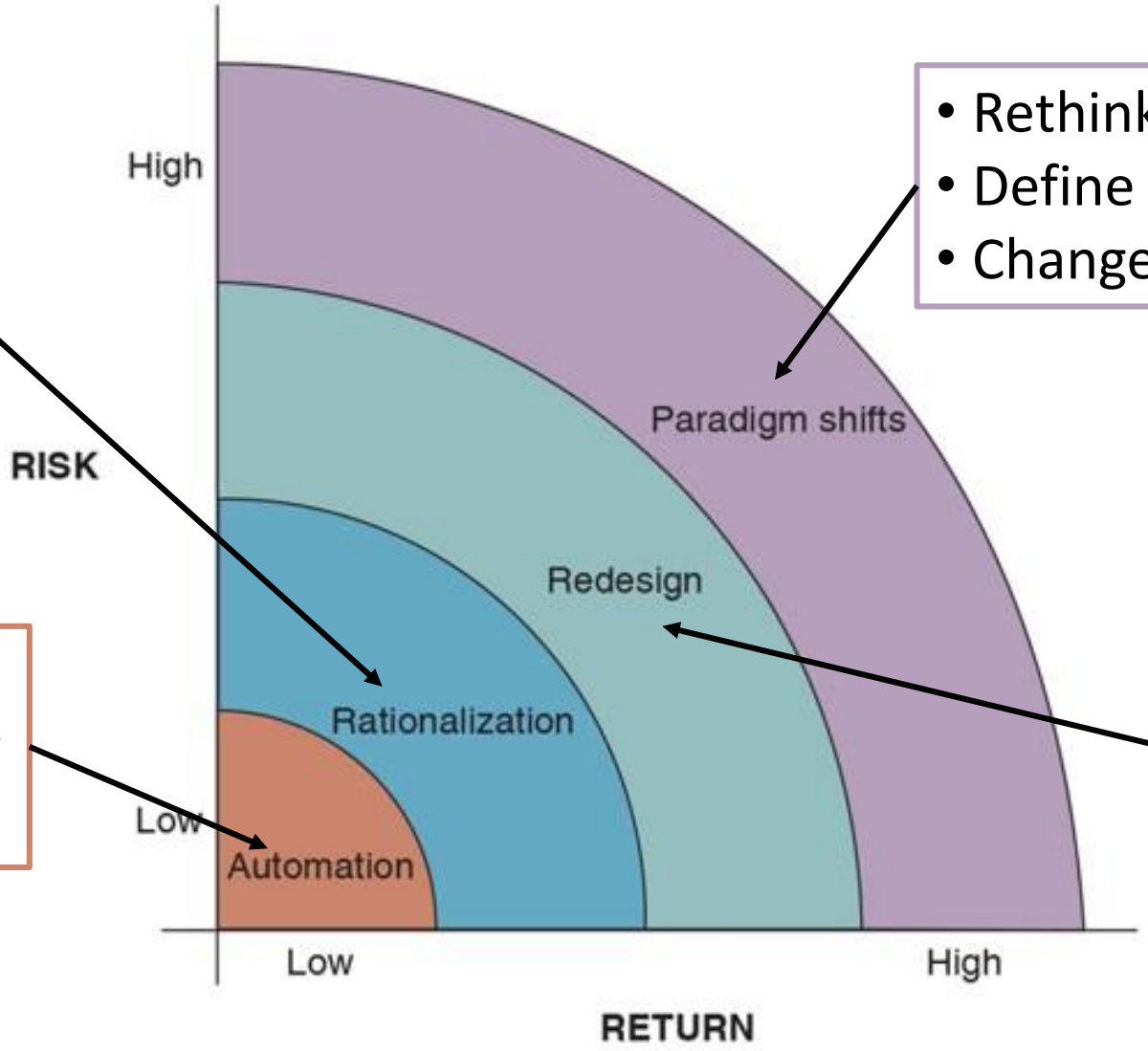
Learning Objectives

1) Explain how new information systems can promote organizational change
2) Understand how IT can be used to manage and redesign business processes
3) Describe the core activities in the systems development process
4) Discuss methods for building and acquiring information systems
5) Describe methods that enable rapid systems development
6) Discuss the increasing democratization of software development
7) Understand the impact of AI on software development
8) Discuss approaches for building mobile applications

IS Drives Organizational Change

The streamlining of standard operating procedures

- Rethink nature of business
- Define new business model
- Change nature of organization



IT helps employees work more efficiently and effectively

Analyze, simplify, and redesign business processes

IS for Managing and Redesign Business Process

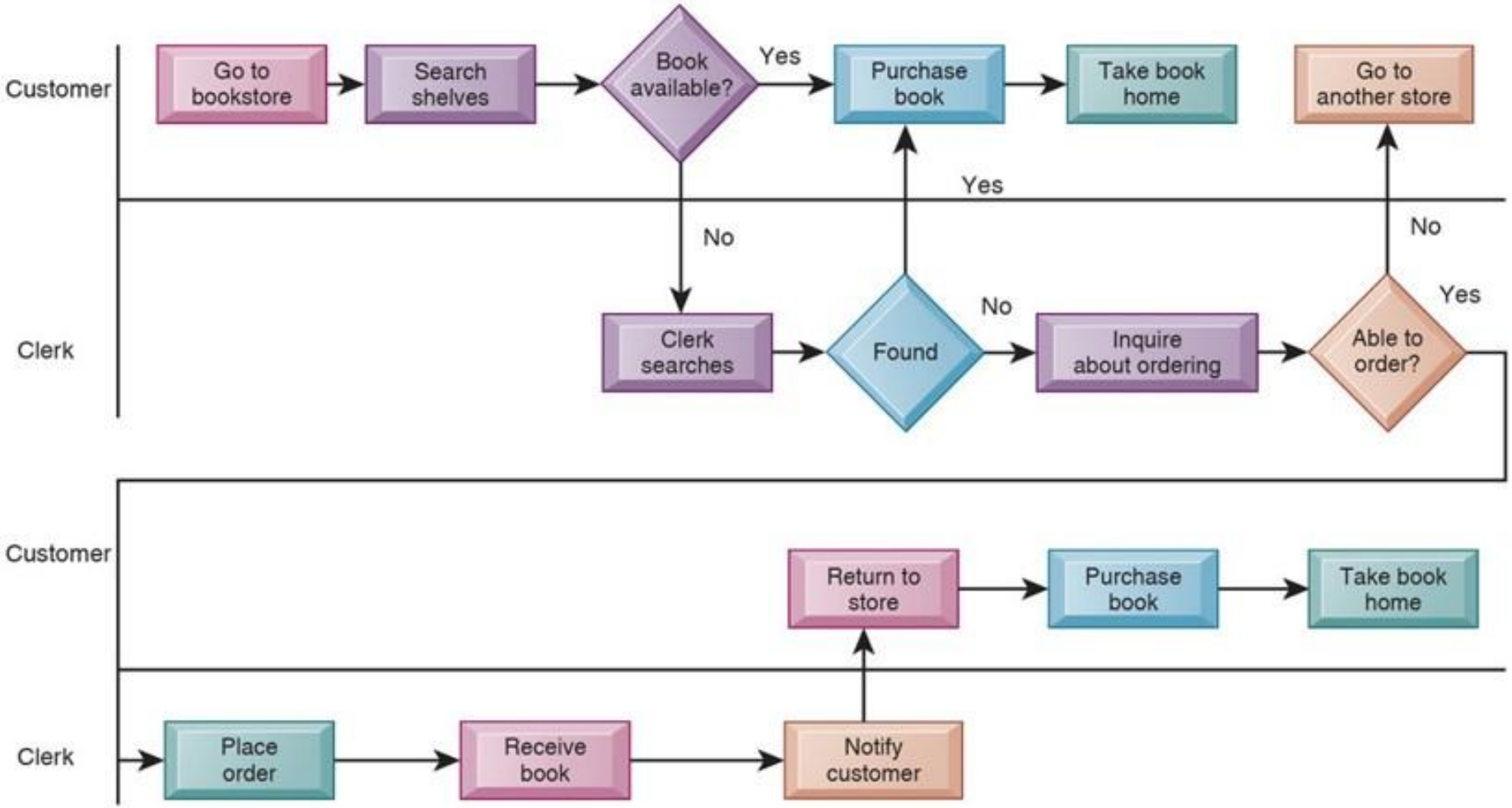
- Business Process Management (BPM) (业务流程管理)
  - A variety of tools and methodologies to
    - analyze existing processes
    - design new processes
    - optimize those processes
- BPM is never concluded because process improvement requires continual change

IS for Managing and Redesign Business Process

Companies practicing business process management go through the following steps:

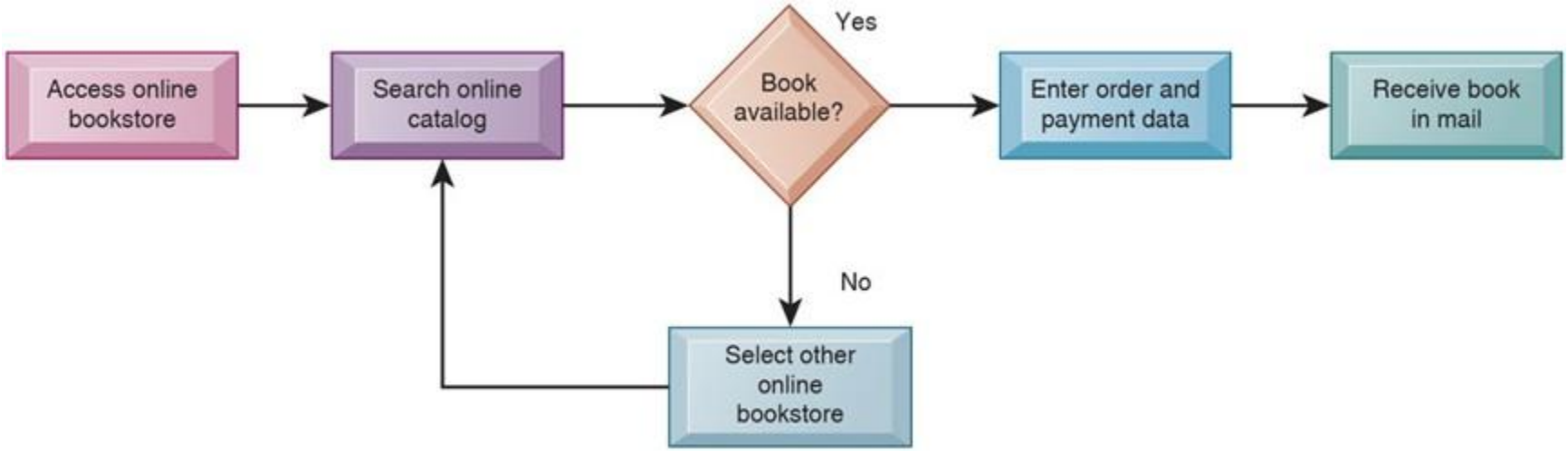
1) Identify processes for change
2) Analyze existing processes
3) Design the new process
4) Implement the new process
5) Continuous measurement

IS for Managing and Redesign Business Process



As-Is Business Process for Purchasing a Book from a **Physical** Bookstore

IS for Managing and Redesign Business Process



Redesigned Process for Purchasing a Book **Online**

Tools for BPM & Process Mining

- Examples: IBM Workflow, Oracle BPM, TIBCO BPM
- Process Mining
  - Definition: Use event logs to uncover inefficiencies and streamline workflows
  - How it works
    - Extract event logs from ERP/CRM systems
    - Builds end-to-end process maps with all real variations
    - Reveals gaps between actual vs. intended workflows
    - Identifies bottlenecks, deviations, and improvement priorities
    - Helps organizations streamline and optimize processes
    - (See case analysis: Celonis case)

Quiz

Which of the following is **not a true statement** about business process management?

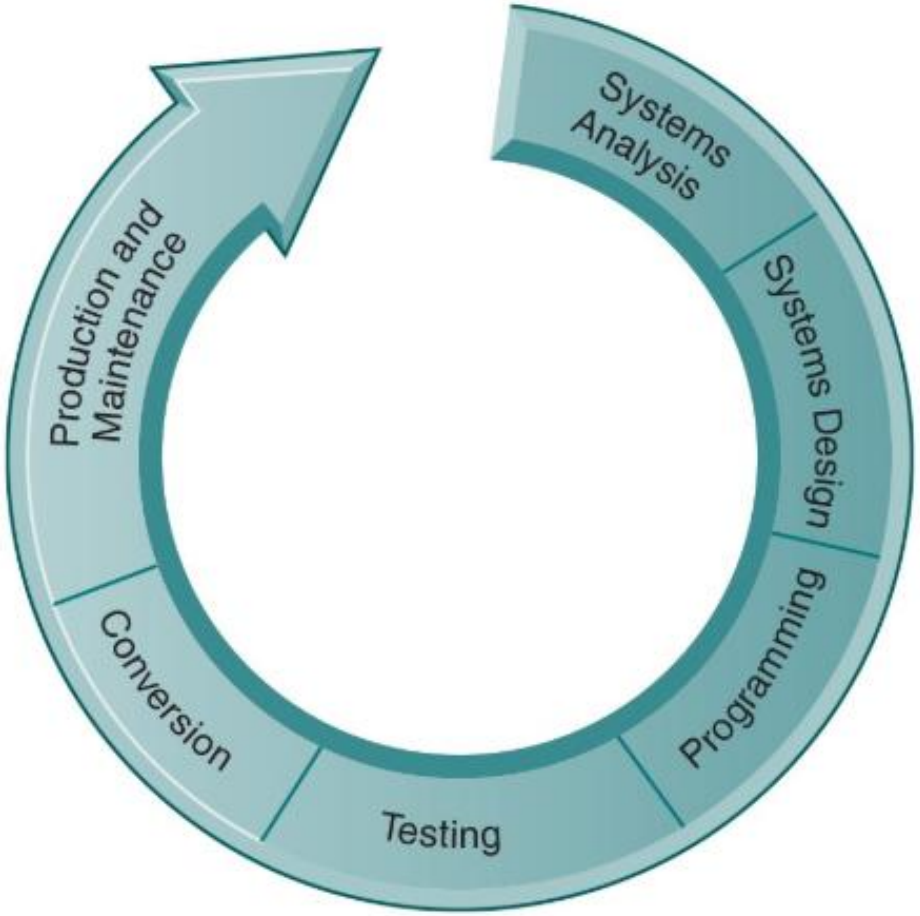
A. The last step in business process management is to implement the new process.

B. Many business process improvements are incremental.

C. Business process management is never finished.

D. Organizational culture is the largest single barrier to successful business process change.

Core Activities in System Development Process (SDP)



Core Activities in SDP – System Analysis

- Purpose
  - Analyze the problem a new IS aims to solve.
  - Define the problem, causes, possible solutions, and information requirements.
- Key Activities
  - Map existing systems, data owners, hardware/software.
  - Identify issues through documents, observations, and user interviews.
  - Conduct a **feasibility study** (financial, technical, organizational).
  - Propose and evaluate alternative solutions.
- Requirements Analysis
  - Determine **who needs what information, when, and how**.
  - Define system objectives and required functions.
  - Poor requirements → major cost overruns or system failure.

Core Activities in SDP – System Design

Output	Processing	Documentation
Medium	Computations	Operations documentation
Content	Program modules	Systems documentation
Timing	Required reports	User documentation
Input	Timing	**Conversion**
Origins	**Manual Procedures**	Data conversion rules
Flow	What activities	Testing method
Data entry	Who performs them	Conversion strategy
User Interface	When	**Training**
Simplicity	How	Training techniques
Efficiency	Where	Training modules
Logic	**Controls**	**Organizational Changes**
Feedback	Input controls (characters, limit, reasonableness)	Task redesign
Errors	Processing controls (consistency, record counts)	Job design
Database Design	Output controls (totals, samples of output)	Process design
Logical data model	Procedural controls (passwords, special forms)	Organization structure design
Volume and speed requirements	**Security**	Reporting relationships
Record specifications	Access controls	
	Catastrophe plans	
	Audit trails	

Core Activities in SDP – Programming, Testing

- Purpose
  - Ensure the system produces the correct results under known conditions.
  - A critical, time-consuming step often underestimated.
- Key Types of Testing
  - Unit Testing: Test individual programs/modules to find errors.
  - System Testing: Test the entire system working together; check performance, storage, load handling, recovery, and procedures.
  - Acceptance Testing: Final user and management verification before production use.

Core Activities in SDP – Conversion

- Purpose
  - Move from the old system to the new system.
- Conversion Strategies
  - Parallel: Run old and new systems together; safest but most expensive.
  - Direct Cutover: Immediate switch; fast but highest risk.
  - Pilot Study: Test new system in one unit first, then roll out organization-wide.
  - Phased Approach: Introduce the new system in stages (by function or unit).
- Key Requirements
  - User training to operate the new system.
  - Clear documentation for technical use and daily operations.
  - Poor training or documentation increases risk of system failure.

Core Activities in SDP – Production & Maintenance

- Production Stage
  - System goes live after conversion.
  - Users and IT review whether it meets original objectives.
  - May include a post-implementation audit.
  - System is fine-tuned based on feedback.
- Maintenance
  - Ongoing changes to hardware, software, documentation, and procedures.
  - Address errors, new requirements, or efficiency improvements.
  - Routine maintenance consumes significant IT resources.
  - Better development practices can reduce long-term maintenance costs.

Core Activities in SDP

Core Activity	Description
Systems analysis	Identify problem(s) Specify solutions Establish information requirements
Systems design	Create design specifications
Programming	Translate design specifications into program code
Testing	Perform unit testing Perform systems testing Perform acceptance testing
Conversion	Plan conversion Prepare documentation Train users and technical staff
Production and maintenance	Operate the system Evaluate the system Modify the system

Methods to Build or Acquire IS

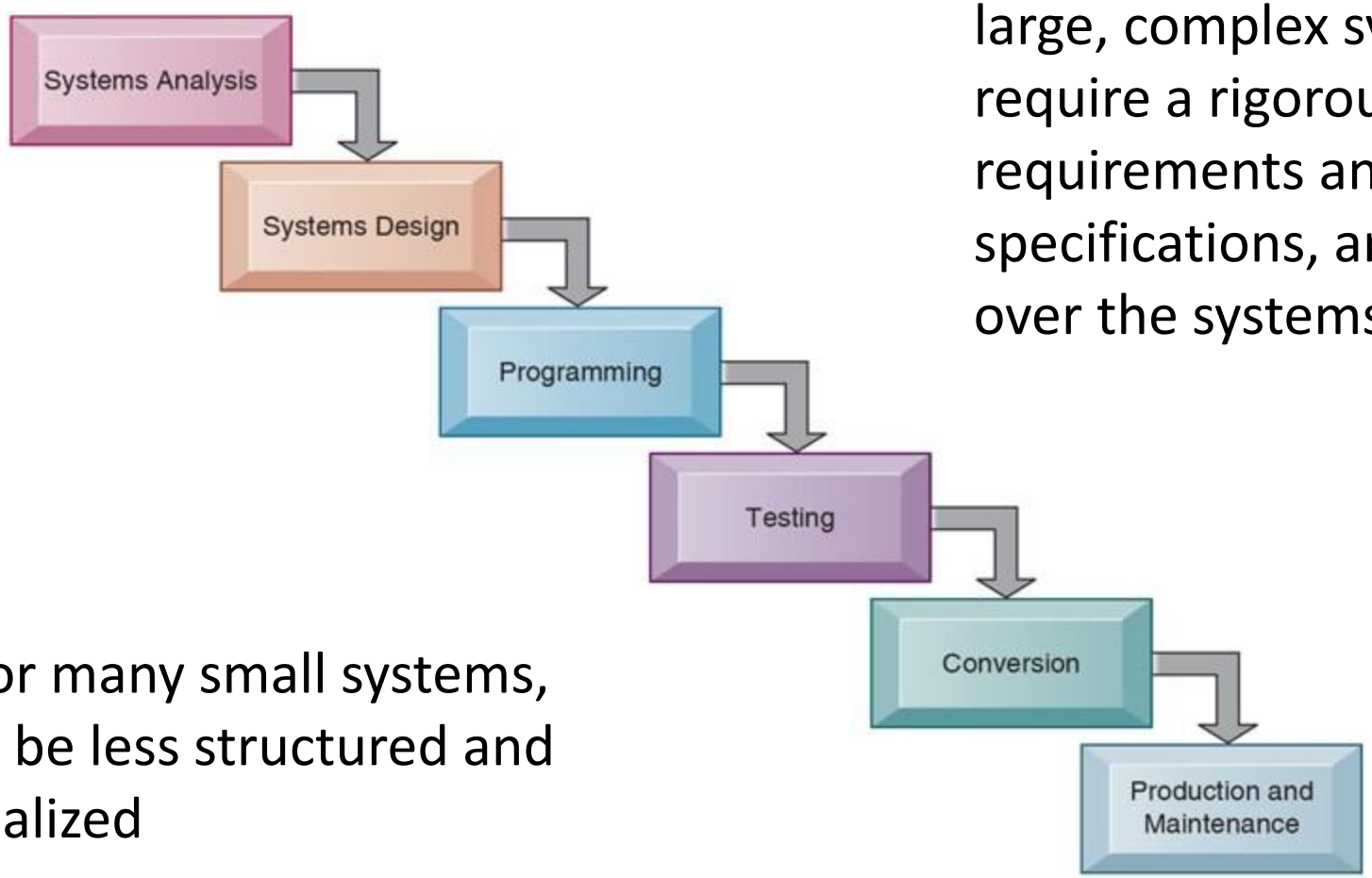
- Traditional Systems Development Life Cycle (SDLC)
- Prototyping
- Application software packages and cloud software services
- Outsourcing

Organizational Problem | Size | Technological Complexity

Traditional Systems Development Life Cycle (SDLC)

Sequential

Waterfall

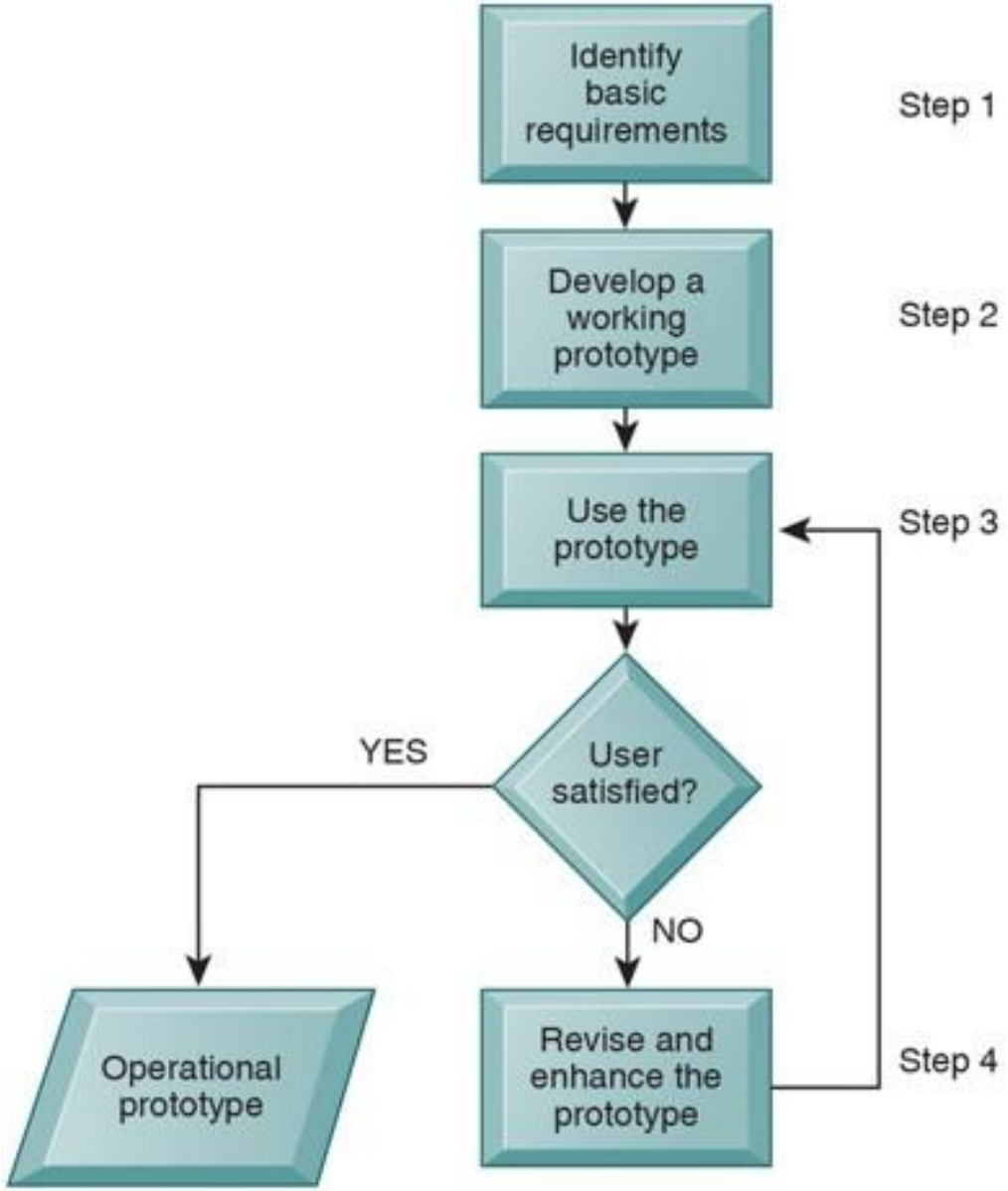


large, complex systems that require a rigorous and formal requirements analysis, predefined specifications, and tight controls over the systems-building process

not suitable for many small systems, which tend to be less structured and more individualized

Prototyping

- Prototyping
  - Building experimental system rapidly and inexpensively for end users to evaluate
- Prototype
  - A working version of an information system but meant to be a preliminary model



Prototyping

- Advantages
  - Useful if some uncertainty in requirements or design solutions
  - Often used for end-user interface design
  - More likely to fulfil the end-user requirements
- Disadvantages
  - May gloss over essential steps
  - May not accommodate large quantities of data or large number of users
    - May not undergo full testing or documentation

Software Package & SaaS

- Application software packages and software services
  - Save time and money
  - Many packages offer customization features
- Evaluation criteria for systems analysis include
  - Functions provided, flexibility, user friendliness, required resources, database requirements, installation and maintenance efforts, documentation, vendor quality, and cost
- Request for proposal (R F P)
  - Detailed list of questions submitted to software vendors
  - Used to evaluate alternative software packages and cloud software services

Outsourcing

1. Cloud and SaaS providers
   - subscribing companies use software and hardware provided by vendors
2. External vendors
   - Hired to design, create software
   - Domestic outsourcing
     - Driven by firm’s need for additional skills, resources, assets
   - Offshore outsourcing
     - Driven by cost-savings

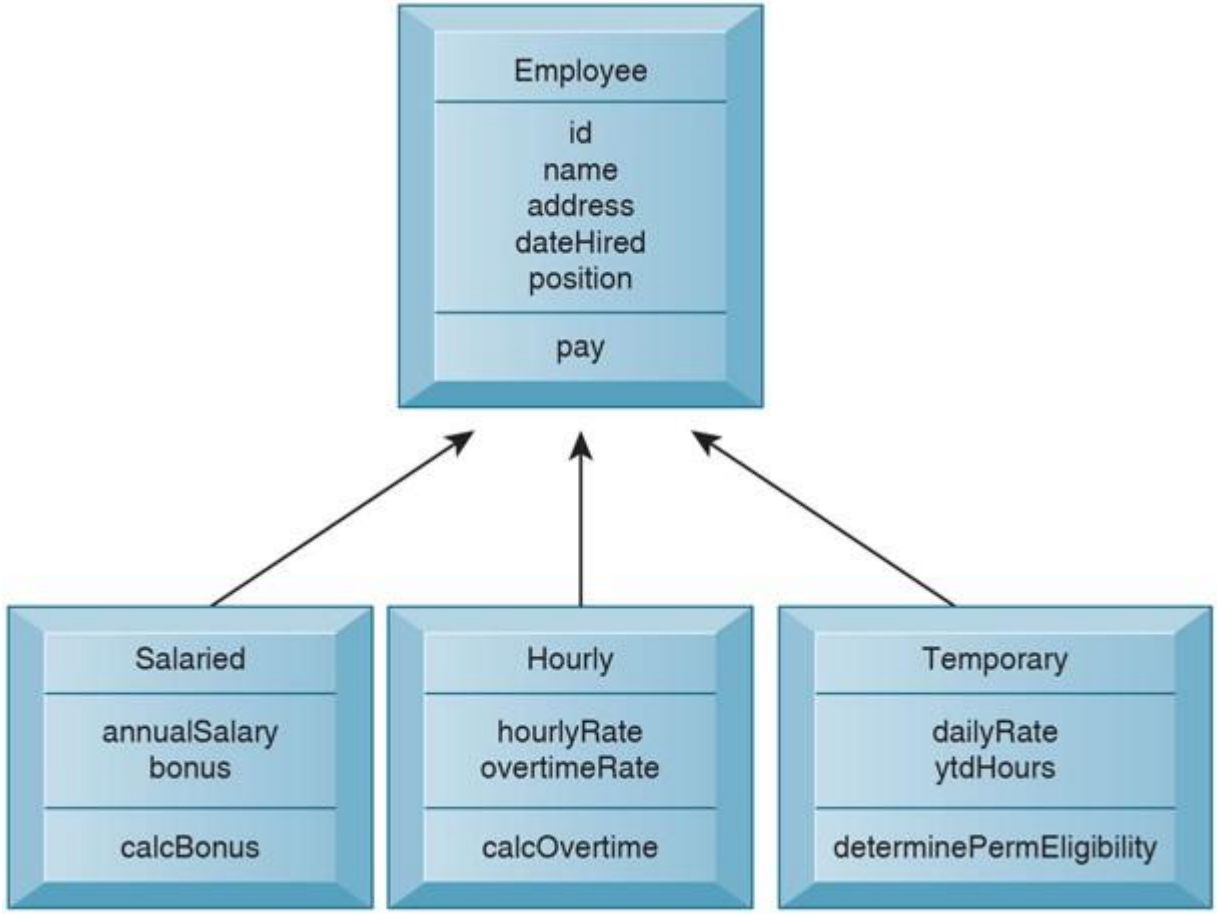
Outsourcing

- Advantages
  - Allows organization flexibility in IT needs
- Disadvantages
  - Hidden costs: identifying and selecting vendor, transitioning to vendor
  - Opening up proprietary business processes to third party

Rapid Systems Development Techniques

- Object-oriented design (OOD)
  - Use an object as the basic unit of systems analysis and designs
- Rapid application development (RAD)
  - The process of creating workable systems in a very short period of time with some flexibility to adapt as a project evolves
- Joint application development (JAD)
  - client, consumer, IT specialists, and stakeholders are all involved in designing and developing a technology application
- Agile development
  - Develops software in short, iterative cycles with continuous feedback and close user involvement.
- DevOps
  - Promotes tight collaboration between development and operations to enable faster, more reliable software delivery.

Class and Inheritance



Democratization of Software Development

- Open-Source Software (OSS)
  - Distributed with source code, allowing programmers to modify it
  - Widely used by major tech companies (Google, Facebook, Microsoft)
  - Pros: free, lower cost, flexible, built-in community
  - Cons: insufficient documentation, compatibility, liability

- Low-Code & No-Code Development
  - Low-code: Faster app development with visual tools and minimal coding
  - No-code: Users create apps using built-in tools without coding
  - Limitations: reduced functionality, risk of unmanaged user-built apps

Impact of AI on Software Development

Function	AI Capabilities
Automated code generation	Generates code by learning from existing patterns; speeds up repetitive tasks.
Automated testing	Creates and runs test cases automatically; improves accuracy and coverage.
Automated bug detection	Detects anomalies using historical bug data; suggests or applies fixes.
Natural language processing (NLP)	Converts user requirements into code specs; supports interactive, conversational development.
Improving IT security	Identifies vulnerabilities in code; recommends secure practices and automates security testing.
Optimizing software performance	Predicts performance issues and resource needs; helps developers tune systems.
Automated documentation	Generates up-to-date documentation from code comments and usage patterns.

Impact of AI on Software Development

Challenges

- AI talent shortage
- Need for large, high-quality data
- High compute costs for training models
- Pretrained models help reduce time and cost (e.g., NVIDIA models)
- AIaaS provides scalable, affordable AI tools (e.g., Google Vertex AI)